

Assignment 4: AI

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Reflection:

For this assignment, I started by creating a test level and figuring out the navmesh feature. Once I had that working, I made a test “guard” with a simple script for only that had the enemy patrolling only, making sure that the pathing and waypoints were working as intended. This took a couple of attempts, but I eventually got one guard moving correctly. After confirming that this worked, I made a state machine visual and began working on implementation, but then I realized that I should probably add player movement first, so I could test the AI logic. I made the player be in first person. I then implemented the state machine, with the guards having a patrolling state, an investigating state (where they go to the player's last known location and look around), and then a chasing state. I also thought that because the guards didn't have faces, it would be easier if they had a vision cone to see where they're facing. I used the Unity Spotlight element and put it on each guard. I also wanted to add a mechanic that would allow the player to “hide”. Originally, we wanted to make the player go invisible, but I found that my approach came with too many bugs. I ended up simplifying this and making it so that crouching would make an enemy stop chasing and investigate.

This was my first time making a more complex game AI using a state machine. I had done both game AI and state machines separately, but never together. I will say that I feel like I got the hang of it pretty quickly. I feel like getting just the patrolling to work in the scene took me longer to understand than just the code for the state machine. A big thing for me was also staying in scope. There were a lot of really fun, crazy things that I wanted to do, but unfortunately did not get around to as I wanted to focus on the technical elements before doing anything artsy. I'm glad I decided to work this way, since we did have to crunch near the deadline in the end.

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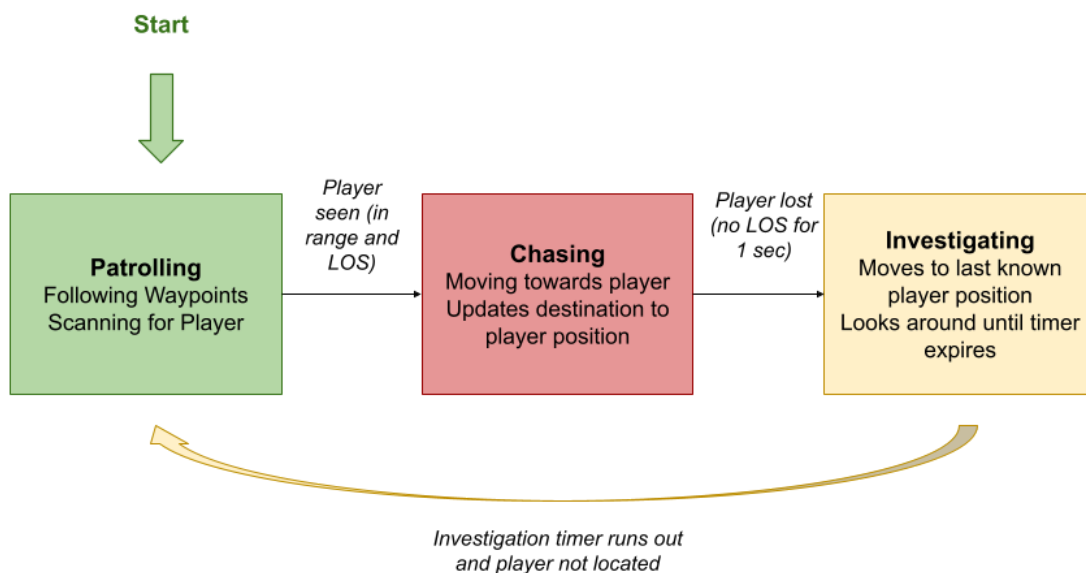
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Reflection: For my part of the project, I worked on creating the main level layout for our stealth game, making the AI pathing, and making the AI interactions like win/lose conditions and made the ending. I made a maze-style map in Unity with a spawn room at the top, an exit room at the

bottom, and enough space in the middle for three NPC guards to patrol. I used walls to shape the level so the player has to move through different paths instead of walking straight to the end. I also added colored floor/path areas to help show where the NPC patrol routes will go.

While working on the level, I learned how important spacing is for AI pathing. I had to make sure the hallways were not too narrow, since the guards need enough room to move with NavMesh. I also learned how to organize the scene with separate walls, floors, and sections so the level is easier to edit later. My goal was to make a layout that supports stealth gameplay by giving the player corners and walls to hide behind while still giving the NPCs clear patrol areas.

Implementation



The guard AI is a finite state machine with 3 states: Patrolling, Investigating, and Chasing. The guards begin in the Patrol state, where they follow waypoint pathing using the NavMeshAgent system. While patrolling, the guard checks if the player is within its vision range/ angle using distance and angle checks as well as raycast.

If a guard detects the player, it transitions into the Chasing state. In this state, the guard constantly updates the NavMesh destination to the player's current location. If an AI is in chasing mode, and the player crouches, the AI will move to an Investigating state.

If the player breaks LOS for a second, the guard enters the Investigating state, goes to the player's last known location, and then looks around. If the player is not found and the Investigating time runs out, the guard goes back to the waypoint location and resumes patrolling.